

BOA JANG

Oculomics · Retinal Imaging Biomarkers · Longitudinal Biomarkers

I develop reliable retinal imaging biomarkers for systemic health and longitudinal disease monitoring.

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EDUCATION

Seoul National University

Ph.D. Candidate in Bioengineering

Seoul, Korea

Mar. 2023 – Present

Peking University

B.S. in Psychological and Cognitive Sciences

Beijing, China

Sep. 2016 – Jul. 2020

RESEARCH EXPERIENCE

Seoul National University Hospital (SNUH)

Medical AI Researcher

Ph.D. Research Collaboration with SNUH

Seoul, Korea

Nov. 2021 – Feb. 2023

Mar. 2023 – Present

Developing AI for ophthalmic imaging in close collaboration with clinicians, spanning nAMD prognosis, retinal foundation models, and imaging biomarkers.

Korea Institute of Science and Technology (KIST)

Research Intern, Brain Science Institute

Seoul, Korea

Aug. 2020 – May 2021

Studied basal ganglia circuits and motor control through in vivo optogenetics, behavioral experiments, and fluorescence imaging.

MANUSCRIPTS UNDER REVIEW / IN REVISION

- [m.5] **B. Jang** et al. Retinal Age Gap as a Biomarker of Biological Aging Reflected by Socioeconomic Status and Lifestyle Behaviors. *GeroScience* (in revision).
- [m.4] **B. Jang** et al. Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures. *AAAI 2027* (under review).
- [m.3] Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning. *Scientific Reports* (in revision).
- [m.2] Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. *Scientific Reports* (in revision).
- [m.1] Age- and Sex-Specific Autism Screening Using Machine Learning Reduction and Conversational Agent Integration. *Translational Psychiatry* (in revision).

PEER-REVIEWED PUBLICATIONS

[†] co-first author

- [j.8] J.R. Song[†], **B. Jang**[†], et al. (2026). Segmentation-Guided Denoising Diffusion Probabilistic Models for Retinal OCT Speckle Noise Reduction with Enhanced Clinical Diagnostic Accuracy. *Scientific Reports* (Accepted).
- [j.7] **B. Jang**, Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. *Scientific Reports*.
- [j.6] **B. Jang**, Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. *Journal of Korean Neurosurgical Society*.
- [j.5] J.H. Lee, S.Y. Lee, **B. Jang**, et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. *Scientific Reports*.
- [j.4] **B. Jang**[†], C.H. Lee[†], et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. *Scientific Reports*.
- [j.3] D.Y. Kim, R. Do, ..., **B. Jang**, et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. *npj Digital Medicine*.
- [j.2] N.Y. Kim[†], **B. Jang**[†], et al. (2024). Differential Diagnosis of Pleural Effusion Using Machine Learning. *Annals of the American Thoracic Society*.
- [j.1] **B. Jang**[†], S.Y. Lee[†], et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. *BMC Ophthalmology*.

PRESENTATIONS

- [o.7] **B. Jang**, G. Lee, J. Choi, and Y.-G. Kim. “Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation.” *2026 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Cheongju, Korea.
- [po.3] **B. Jang**, K.H. Bae, and Y.-G. Kim. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)*, Korea.
- [o.6] **B. Jang** et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *MICCAI 2025 AMAI Workshop*, Daejeon, Korea. **Best Abstract Award – Honorable Mention.**
- [o.5] **B. Jang** et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *2025 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Busan, Korea. **Best Paper Award.**
- [o.4] **B. Jang** et al. “Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology.” *The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)*, Seoul, Korea.
- [o.3] **B. Jang** and Y.-G. Kim. “Foundation Models in Healthcare: Development, Necessity and Implementation Strategies.” *2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)*, Las Vegas, NV, USA.
- [o.2] **B. Jang** et al. “The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists.” *2023 Fall Conference of The Korean Society of Medical Informatics (KoSMI)*, Bundang, Korea. **Best Paper Award.**
- [po.2] **B. Jang** et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” *10th Workshop on Ophthalmic Medical Image Analysis (OMIA-X), 2023 in International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Vancouver, Canada.
- [po.1] **B. Jang**, Y.-G. Kim, and E.K. Lee. “Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning.” *2023 American Society of Retina Specialists (ASRS)*, Seattle, WA, USA. **Poster Award Winner.**
- [o.1] **B. Jang** et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” *2023 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Daegu, Korea. **Best Paper Award.**

AWARDS

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|-------|---|------------|
| [a.5] | AI Seoul Tech Graduate Scholarship, Seoul Future Talent Foundation | Nov. 2025 |
| [a.4] | Hana Bank President’s Award, 2024 Digital Wellness Competition | Nov. 2024 |
| [a.3] | 2nd Place, Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6 | Dec. 2023 |
| [a.2] | 3rd Place, Digital Pathology AI Hackathon, CODiPAI Challenge | Oct. 2023 |
| [a.1] | SNUAIMED Excellence Scholarship | 2023, 2024 |

TEACHING & MENTORING

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|-------|---|------------|
| [t.4] | Teaching Assistant, Korea Clinical Datathon (KCD) 2025 | Oct. 2025 |
| [t.3] | Student Ambassador, MICCAI 2025 AMAI Workshop | Sep. 2025 |
| [t.2] | Teaching Assistant, Clinical and Healthcare AI Training Programs | 2024, 2025 |
| [t.1] | Teaching Assistant, NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022 | May 2022 |

PATENTS

- [p.2] Y.-G. Kim, . . . , **B. Jang**. Controllable Normal-to-Disease Fundus Translation via Factorial Guidance. Korea Patent Application No. 10-2026-0119988, filed June 2026. (*Pending*)
- [p.1] H.-S. Kim, . . . , Y.-G. Kim, **B. Jang**. Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data. Korea Patent No. 10-2953173, registered April 2026. (*Granted*)

REFERENCES

Prof. Jin-Wook Choi, PhD (jinchoi@snu.ac.kr)
Prof. Young-Gon Kim, PhD (younggon2.kim@gmail.com)
Prof. Hyuk Jin Choi, MD, PhD (docchoi@hanmail.net)